

TORBAT-HEYDARIEH, Iran, Islamic Republic of

WMO: 407620

Lat: 35.3315N

Lon: 59.2059E

Elev: 1451

StdP: 85.07

Time Zone: 3.50 (IRN)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-9.7	-6.9	-18.5	0.9	-2.0	-15.4	1.2	-0.1	9.2	2.2	7.9	1.3	1.7	330	0.489

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	35.1	16.6	33.9	16.2	32.8	15.9	18.7	31.8	17.8	31.1	17.0	30.5	4.9	90

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
13.5	11.5	26.8	12.2	10.5	25.4	11.0	9.8	24.3	59.5	32.0	56.3	31.1	53.5	30.5	23.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature							
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years	
(a)	(b)	(c)	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max
9.9	8.4	7.5	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
			-12.9	37.7	3.0	1.1	-15.0	38.5	-16.8	39.2	-18.5	39.8	-20.7	40.6
			-13.4	19.3	2.9	1.6	-15.5	20.5	-17.2	21.4	-18.8	22.3	-20.9	23.4
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Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	14.7	1.3	3.5	8.7	14.6	20.6	25.3	27.2	25.7	21.4	15.3	8.2	3.4	(6)
(7)		DBStd	9.69	4.07	4.15	4.34	4.34	3.49	2.70	2.21	2.24	3.08	3.78	3.88	3.32	(7)
(8)		HDD10.0	826	268	183	76	12	0	0	0	0	4	78	205		(8)
(9)		HDD18.3	2265	527	414	299	123	17	1	0	9	107	305	462		(9)
(10)		CDD10.0	2527	0	2	36	151	328	458	532	486	341	169	24	1	(10)
(11)		CDD18.3	923	0	0	1	12	87	208	274	228	100	14	0	0	(11)
(12)		CDH23.3	10102	0	0	9	120	840	2299	3177	2419	1057	181	1	0	(12)
(13)		CDH26.7	4283	0	0	1	21	272	1021	1526	1048	364	31	0	0	(13)
(14)	Wind	WSAvg	2.9	1.6	2.1	2.6	3.0	3.4	3.9	4.6	4.1	2.9	2.4	1.9	1.6	(14)
(15)	Precipitation	PrecAvg	264	37	43	58	40	25	4	1	2	7	18	28		(15)
(16)		PrecMax	424	82	133	144	90	86	41	22	3	18	51	44	94	(16)
(17)		PrecMin	101	7	2	4	1	1	0	0	0	0	0	0	0	(17)
(18)		PrecStd	83	20	33	38	26	20	9	4	1	4	12	13	24	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	14.1	17.5	24.5	29.3	33.2	36.2	37.1	35.6	33.8	29.6	22.5	16.7	(19)
(20)			MCWB	6.1	7.6	11.2	13.0	15.0	16.7	17.3	16.8	15.0	13.3	10.7	6.9	(20)
(21)		2%	DB	11.9	14.3	21.0	26.4	31.2	34.5	35.5	34.0	31.9	27.2	19.7	13.5	(21)
(22)			MCWB	5.1	6.5	9.9	12.8	14.7	16.1	16.9	16.2	14.4	12.6	9.2	5.6	(22)
(23)		5%	DB	9.8	12.3	18.8	24.5	29.7	33.1	34.2	32.7	30.3	25.4	17.7	11.7	(23)
(24)			MCWB	4.1	5.6	9.0	12.4	14.3	15.9	16.6	15.8	14.1	11.9	8.2	4.7	(24)
(25)		10%	DB	7.9	10.5	16.6	22.5	27.9	31.8	33.0	31.5	28.8	23.3	15.7	9.7	(25)
(26)			MCWB	3.1	4.9	8.1	11.5	14.0	15.5	16.3	15.3	13.7	11.1	7.3	3.9	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	7.7	9.3	12.4	15.0	17.1	19.5	20.2	19.6	17.5	14.2	11.8	7.7	(27)
(28)			MCDWB	11.4	13.4	19.3	25.0	28.7	31.8	33.3	32.4	29.8	26.4	19.5	13.8	(28)
(29)		2%	WB	6.1	7.9	11.1	13.9	15.9	18.2	19.0	18.1	16.1	13.2	10.2	6.5	(29)
(30)			MCDWB	10.2	11.8	18.1	23.9	28.2	31.2	32.3	31.4	29.0	25.1	17.8	11.8	(30)
(31)		5%	WB	4.9	6.7	10.0	13.1	15.2	17.2	18.1	17.2	15.1	12.5	9.1	5.5	(31)
(32)			MCDWB	8.8	11.0	16.8	22.5	26.9	30.3	31.6	30.7	28.2	23.9	15.7	10.3	(32)
(33)		10%	WB	3.6	5.6	8.9	12.2	14.5	16.3	17.2	16.3	14.2	11.6	8.0	4.5	(33)
(34)			MCDWB	7.3	9.9	15.4	20.7	25.9	29.4	31.0	29.7	27.3	22.3	14.1	8.8	(34)
(35)	Mean Daily Temperature Range	MDBR	10.2	10.5	11.5	11.6	11.9	11.6	10.7	11.3	13.8	13.8	11.8	10.7	(35)	
(36)		5% DB	MCDBR	12.3	13.1	14.4	13.7	13.7	13.0	12.3	12.6	15.0	15.1	14.1	13.5	(36)
(37)			MCWBR	7.2	7.1	6.8	5.6	5.3	5.4	5.1	5.6	6.9	7.3	7.7	7.9	(37)
(38)		5% WB	MCDWB	10.8	10.8	12.1	12.0	12.3	11.8	11.5	11.4	13.7	13.9	12.0	11.5	(38)
(39)			MCWBR	6.7	6.4	6.0	5.1	5.2	5.7	5.8	5.7	6.9	6.6	6.7	7.0	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.306	0.343	0.409	0.450	0.466	0.442	0.434	0.395	0.360	0.375	0.341	0.303	(40)	
(41)		taud	2.379	2.255	2.047	1.976	1.954	2.027	2.057	2.170	2.252	2.179	2.279	2.408	(41)	
(42)		Ebn at Noon	893	891	854	837	827	845	848	877	894	840	836	874	(42)	
(43)		Edh at Noon	92	118	158	179	186	172	166	146	127	125	101	85	(43)	
(44)	All-Sky Solar Radiation	RadAvg	2.75	3.60	4.78	6.02	7.15	8.16	8.03	7.48	6.30	4.70	3.28	2.62	(44)	
(45)		RadStd	0.28	0.39	0.31	0.42	0.37	0.23	0.13	0.20	0.15	0.18	0.23	0.25	(45)	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
(46)	Station Only	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
		+0.50	N/A	N/A	+0.88	-0.77	N/A	N/A	N/A	N/A	N/A
(47)	Regional (4 neighbors)	+0.40	N/A	-1.79	+0.45	-1.18	-1.15	N/A	N/A	N/A	+89

Nomenclature: See separate page